NCEA Level 2 Mathematics

91261 · Apply algebraic methods in solving problems (Algebra)

Excellence-level practice questions

SOLUTIONS

About these solutions

- These are **full worked solutions** to the 91261 practice excellence questions. Where a question can be done more than one way, an **alternate solution** is also shown.
- They are **not** an official NZQA assessment schedule, and no marks or grades are given. However, solving these levels of questions in the exam should award you an excellence mark.

How to use them

- Attempt each question properly first, then check your final answer against the answer schedule.
- Only read a solution once you're stuck or have finished, and don't just copy it. Close it and try the question again on your own.
- Your method doesn't have to match the one shown, as long as it is correct and fully explained.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

Assessment Schedule – Practice Excellence Questions

Level 2 Mathematics: Algebra (91261)

Evidence Statement

Q	Solution
1	$3^{2x+1} = 3\,(3^x)^2$. Let $u = 3^x$: $3u^2 - 28u + 9 = 0$ $(3u - 1)(u - 9) = 0$ $u = \frac{1}{3}$ or $u = 9$ $3^x = \frac{1}{3} \Rightarrow x = -1$ $\quad$ $3^x = 9 \Rightarrow x = 2$
2	Let $u = x^2 - 3x$: $u^2 - 2u - 8 = 0$ $(u - 4)(u + 2) = 0$ $u = 4$: $x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0 \Rightarrow x = 4$ or $x = -1$ $u = -2$: $x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0 \Rightarrow x = 1$ or $x = 2$ $x = -1,\ 1,\ 2,\ 4$
3	Domain: need $x + 2 > 0$ and $x - 4 > 0$, so $x > 4$. Change the base 4 log to base 2: $\log_4(x + 2) = \dfrac{\log_2(x + 2)}{\log_2 4} = \frac{1}{2}\log_2(x + 2)$ $\log_2(x + 2) - \log_2(x - 4) = 2$ $\log_2 \dfrac{x + 2}{x - 4} = 2$ $x + 2 = 4(x - 4)$ $x = 6$ $x = 6 > 4$, so it is valid. Check: $2\log_4 8 - \log_2 2 = 2(1.5) - 1 = 2$ ✓ (Any value $x \le 4$ would be rejected, as the log of a non-positive number is undefined.) **Alternate solution:** Change the base 2 log to base 4: $\log_2(x - 4) = 2\log_4(x - 4)$ $2\log_4(x + 2) - 2\log_4(x - 4) = 2$ $\log_4 \dfrac{x + 2}{x - 4} = 1$ $x + 2 = 4x - 16 \Rightarrow x = 6$
4	$\log_a x = 3 \Rightarrow x = a^3 \Rightarrow a = x^{\frac{1}{3}}$ $\quad$ $\log_b x = 4 \Rightarrow x = b^4 \Rightarrow b = x^{\frac{1}{4}}$ $ab = x^{\frac{1}{3}} \times x^{\frac{1}{4}} = x^{\frac{7}{12}}$ $x = (ab)^{\frac{12}{7}}$ $\log_{ab} x = \dfrac{12}{7}$ **Alternate solution:** Using change of base, $\log_x a = \dfrac{1}{\log_a x} = \dfrac{1}{3}$ and $\log_x b = \dfrac{1}{4}$. $\log_x(ab) = \log_x a + \log_x b = \dfrac{1}{3} + \dfrac{1}{4} = \dfrac{7}{12}$ $\log_{ab} x = \dfrac{1}{\log_x(ab)} = \dfrac{12}{7}$

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Q	Solution
5	$\dfrac{3^x}{3} = 5 \times 2 \times 2^x$ $\dfrac{3^x}{2^x} = 30$ $\left(\frac{3}{2}\right)^x = 30$ $x \log 1.5 = \log 30$ $x = \dfrac{\log 30}{\log 1.5} \ (\approx 8.39)$ **Alternate solution:** Take logs of both sides first: $(x-1)\log 3 = \log 5 + (x+1)\log 2$ $x \log 3 - x \log 2 = \log 3 + \log 5 + \log 2$ $x \log \frac{3}{2} = \log 30 \ \Rightarrow \ x = \dfrac{\log 30}{\log 1.5}$
6	$Ar^7 = 25\,000 \quad \text{and} \quad Ar^3 = 4000$ Divide: $r^4 = 6.25 \ \Rightarrow \ r = 6.25^{\frac{1}{4}} \approx 1.5811$ $A = \dfrac{4000}{r^3} \approx 1011.9$ $1011.9 \times 1.5811^t > 100\,000$ $1.5811^t > 98.82$ $t > \dfrac{\log 98.82}{\log 1.5811} \approx 10.03$ The number of bacteria first exceeds 100 000 after about 10.0 hours. **Alternate solution:** Start the count from $t = 3$, so $N = 4000\,r^{t-3}$ (no need to find A): $4000\,r^{t-3} > 100\,000$ $r^{t-3} > 25$ $t - 3 > \dfrac{\log 25}{\log 1.5811} \approx 7.03 \ \Rightarrow \ t > 10.03$
7	After n years: Aroha $= 5000(1.042)^n$, Ben $= 6500(1.029)^n$ $5000(1.042)^n > 6500(1.029)^n$ $\left(\dfrac{1.042}{1.029}\right)^n > 1.3$ $n \log\left(\dfrac{1.042}{1.029}\right) > \log 1.3$ $n > 20.9$ Aroha's investment is first worth more after 21 complete years. **Alternate solution:** Take logs of both sides straight away: $\log 5000 + n \log 1.042 > \log 6500 + n \log 1.029$ $n(\log 1.042 - \log 1.029) > \log 6500 - \log 5000$ $n > \dfrac{\log 1.3}{\log 1.042 - \log 1.029} \approx 20.9 \ \Rightarrow \ 21 \text{ years}$

Q	Solution
8	At $d = 3$: $0.6\,I_0 = I_0\,k^3 \Rightarrow k = 0.6^{\frac{1}{3}} \approx 0.8434$ $I_0\,k^d \geq 0.1\,I_0$ $0.8434^d \geq 0.1$ $d \log 0.8434 \geq \log 0.1$ $d \leq \dfrac{\log 0.1}{\log 0.8434}$ (inequality reverses, since $\log 0.8434 < 0$) $d \leq 13.5$ The greatest depth is about 13.5 m. **Alternate solution:** Without finding k: $I = I_0\,(0.6)^{\frac{d}{3}}$ $0.6^{\frac{d}{3}} \geq 0.1$ $\dfrac{d}{3} \leq \dfrac{\log 0.1}{\log 0.6} \approx 4.508 \Rightarrow d \leq 13.5$ m
9	Since $k \neq 0$, the equation is a quadratic. $\Delta = (k+3)^2 - 4(k)(1)$ $= k^2 + 6k + 9 - 4k$ $= k^2 + 2k + 9$ $= (k+1)^2 + 8$ $(k+1)^2 \geq 0$, so $\Delta \geq 8 > 0$ for all k. Therefore there are always two distinct real roots. **Alternate solution:** Show $k^2 + 2k + 9 > 0$ using its own discriminant: $2^2 - 4(1)(9) = -32 < 0$, so $k^2 + 2k + 9$ has no real zeros. Its leading coefficient is positive, so it is always positive, i.e. $\Delta > 0$ for all k.
10	$x^2 - (a+b)x + ab - k^2 = 0$ $\Delta = (a+b)^2 - 4(ab - k^2)$ $= a^2 + 2ab + b^2 - 4ab + 4k^2$ $= (a-b)^2 + 4k^2$ $(a-b)^2 \geq 0$ and $4k^2 \geq 0$, so $\Delta \geq 0$: the equation always has real solutions. Exactly one solution when $\Delta = 0$, i.e. $(a-b)^2 = 0$ and $k^2 = 0$: $a = b$ **and** $k = 0$. **Alternate solution:** The graph of $y = (x-a)(x-b)$ is a parabola opening upwards that crosses or touches the x-axis at $x = a$ and $x = b$, so its minimum value is ≤ 0. Since $k^2 \geq 0$, the horizontal line $y = k^2$ is never below the minimum, so it always meets the parabola: always real solutions. There is exactly one solution only if the line touches the vertex. This needs k^2 = minimum value ≤ 0, so $k = 0$ and the minimum is 0, which means the parabola touches the x-axis: $a = b$.

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Q	Solution
11	$x^2 + 3x + 3 = mx - 1$ $x^2 + (3 - m)x + 4 = 0$ Tangent, so $\Delta = 0$: $(3 - m)^2 - 16 = 0$ $3 - m = \pm 4$ $m = -1$ or $m = 7$ Repeated root $x = \dfrac{m-3}{2}$: $m = 7$: $x = 2$, $y = 13$ $\Rightarrow (2, 13)$ $m = -1$: $x = -2$, $y = 1$ $\Rightarrow (-2, 1)$ **Alternate solution:** Using calculus: the gradient of the curve is $2x + 3 = m$. The point is on both the line and the curve: $x^2 + 3x + 3 = (2x + 3)x - 1$ $x^2 = 4 \Rightarrow x = \pm 2$ $x = 2$: $m = 7$, point $(2, 13)$; $x = -2$: $m = -1$, point $(-2, 1)$
12	Two distinct real roots: $m \neq 0$ and $\Delta > 0$ $16 - 4m(m - 3) > 0$ $m^2 - 3m - 4 < 0$ $(m - 4)(m + 1) < 0 \Rightarrow -1 < m < 4$ Both roots negative: Sum of roots $= -\dfrac{4}{m} < 0 \Rightarrow m > 0$ Product of roots $= \dfrac{m-3}{m} > 0$, and with $m > 0$ this gives $m > 3$ Combining: $3 < m < 4$ **Alternate solution:** (Without sum and product of roots.) Let $f(x) = mx^2 + 4x + (m - 3)$. The axis of symmetry is $x = -\dfrac{2}{m}$. If $m < 0$, the axis is at a positive x-value, and the roots are symmetric about it, so at least one root is positive. So $m > 0$, the parabola opens upwards and the axis is at a negative x-value. With both roots negative, $x = 0$ lies to the right of both roots, so $f(0) > 0$: $m - 3 > 0 \Rightarrow m > 3$. Combining with $-1 < m < 4$ from $\Delta > 0$: $3 < m < 4$

Q	Solution
13	Roots: $x = \dfrac{-b \pm \sqrt{b^2 - 40}}{2}$ Difference between the roots: $\dfrac{-b + \sqrt{b^2 - 40}}{2} - \dfrac{-b - \sqrt{b^2 - 40}}{2} = \sqrt{b^2 - 40} = 3$ $b^2 - 40 = 9$ $b = \pm 7$ Check: $x^2 + 7x + 10 = (x + 2)(x + 5)$, roots $-2, -5$ differ by 3 ✓ **Alternate solution:** Let the roots be α and $\alpha + 3$: $x^2 + bx + 10 = (x - \alpha)(x - \alpha - 3) = x^2 - (2\alpha + 3)x + \alpha(\alpha + 3)$ $\alpha^2 + 3\alpha - 10 = 0 \Rightarrow (\alpha + 5)(\alpha - 2) = 0$ $\alpha = 2: \ b = -(2\alpha + 3) = -7 \qquad \alpha = -5: \ b = 7$
14	Let the roots be $2r$ and $3r$. Then $ax^2 + bx + c = a(x - 2r)(x - 3r)$ $= ax^2 - 5ar\,x + 6ar^2$ Equating coefficients: $b = -5ar, \ c = 6ar^2$ $6b^2 = 6 \times 25a^2r^2 = 150a^2r^2$ $25ac = 25a \times 6ar^2 = 150a^2r^2$ $\therefore \ 6b^2 = 25ac$ **Alternate solution:** Using sum and product of roots: $2r + 3r = -\dfrac{b}{a}$ and $(2r)(3r) = \dfrac{c}{a}$ $r = -\dfrac{b}{5a}$ $6\left(\dfrac{b^2}{25a^2}\right) = \dfrac{c}{a}$ $6b^2 = 25ac$
15	$x \neq 1$ and $x \neq -1$. Multiply both sides by $(x - 1)(x + 1)$: $(x + 1) + (x - 1) = k(x^2 - 1)$ $kx^2 - 2x - k = 0$ **Case** $k = 0$**:** $-2x = 0 \Rightarrow x = 0$, which is allowed. Exactly one solution. **Case** $k \neq 0$**:** the equation is a quadratic with $\Delta = (-2)^2 - 4(k)(-k) = 4 + 4k^2 > 0$ so there are two distinct real roots. Neither root can be ± 1: $x = 1: \ k - 2 - k = -2 \neq 0; \quad x = -1: \ k + 2 - k = 2 \neq 0$ So for every $k \neq 0$ there are exactly two solutions. Therefore the equation has exactly one solution only when $k = 0$.

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Q	Solution
16	$x^2 - 16 = (x-4)(x+4)$, so $\dfrac{x-3}{x-4} = \dfrac{(x-3)(x+4)}{(x-4)(x+4)} = \dfrac{x^2+x-12}{x^2-16}$ Comparing numerators: $x^2 + kx - 12 = x^2 + x - 12 \Rightarrow k = 1$ Not valid when a denominator is zero: $x = 4$ or $x = -4$ **Alternate solution:** Cross-multiply: $(x^2 + kx - 12)(x-4) = (x-3)(x^2-16)$ $x^3 + (k-4)x^2 - (4k+12)x + 48 = x^3 - 3x^2 - 16x + 48$ x^2 terms: $k - 4 = -3 \Rightarrow k = 1$ (check x terms: $-(4+12) = -16$ ✓)
17	Let n be the number of friends in the original group. $\dfrac{360}{n} - \dfrac{360}{n+4} = 15$ $360(n+4) - 360n = 15n(n+4)$ $1440 = 15n^2 + 60n$ $n^2 + 4n - 96 = 0$ $(n+12)(n-8) = 0$ $n > 0$, so $n = 8$. Check: \$45 each, then \$30 each with 12 people ✓ **Alternate solution:** Let the original cost per person be \$$c$: $nc = 360$ and $(n+4)(c-15) = 360$ $nc - 15n + 4c - 60 = 360 \Rightarrow 4c = 15n + 60$ $n\left(\dfrac{15n+60}{4}\right) = 360 \Rightarrow n^2 + 4n - 96 = 0 \Rightarrow n = 8$
18	Let the speed of the boat in still water be v km h^{-1} ($v > 2$). Upstream speed $v - 2$, downstream speed $v + 2$. Time = $\dfrac{\text{distance}}{\text{speed}}$: $\dfrac{24}{v-2} + \dfrac{24}{v+2} = 5$ $24(v+2) + 24(v-2) = 5(v^2-4)$ $5v^2 - 48v - 20 = 0$ $(5v+2)(v-10) = 0$ $v > 2$, so $v = 10$ km h^{-1}. Check: $3 + 2 = 5$ hours ✓

Q	Solution
19	Put the origin at ground level under the top of the arch: $y = 12 - ax^2$, $y = 0$ when $x = 15$ $0 = 12 - 225a \Rightarrow a = \dfrac{12}{225} = \dfrac{4}{75}$ Top edge of the sign is $7 + 2 = 9$ m above the ground: $9 = 12 - \dfrac{4}{75}x^2$ $x^2 = 56.25 \Rightarrow x = \pm 7.5$ Width of the sign $= 7.5 - (-7.5) = 15$ m **Alternate solution:** Put the origin at one foot of the arch, so $y = ax(x - 30)$. The vertex is $(15, 12)$: $12 = a(15)(-15) \Rightarrow a = -\dfrac{4}{75}$ $9 = -\dfrac{4}{75}x(x - 30) \Rightarrow x^2 - 30x + 168.75 = 0$ $x = \dfrac{30 \pm \sqrt{900 - 675}}{2} = 7.5$ or 22.5 Width $= 22.5 - 7.5 = 15$ m
20	Outer rectangle: length $12 + 2(2x) = 12 + 4x$, width $8 + 2x$ Area of path = area of garden, so the outer area is $2 \times 96 = 192$: $(12 + 4x)(8 + 2x) = 192$ $8x^2 + 56x + 96 = 192$ $x^2 + 7x - 12 = 0$ $x = \dfrac{-7 \pm \sqrt{49 + 48}}{2} = \dfrac{-7 \pm \sqrt{97}}{2}$ $x > 0$, so $x = \dfrac{-7 + \sqrt{97}}{2} \approx 1.42$ m **Alternate solution:** Add up the pieces of the path. Two strips along the long sides: $2 \times 12x = 24x$. Two strips along the short sides (including the corners): $2 \times 2x(8 + 2x) = 32x + 8x^2$. $8x^2 + 56x = 96$ $x^2 + 7x - 12 = 0 \Rightarrow x \approx 1.42$ m
21	$2^x \times 2^{2y} = 2^6 \Rightarrow x + 2y = 6$ $\dfrac{3^x}{3^{2y}} = 3^{-3} \Rightarrow x - 2y = -3$ Adding: $2x = 3 \Rightarrow x = \dfrac{3}{2}$. Then $2y = 6 - \dfrac{3}{2} \Rightarrow y = \dfrac{9}{4}$

Q	Solution
22	$\log_4 y = \dfrac{\log_2 y}{\log_2 4} = \frac{1}{2}\log_2 y$ $\log_2 x + \frac{1}{2}\log_2 y = 4$ $\log_2\left(x\sqrt{y}\right) = 4 \Rightarrow x\sqrt{y} = 16$ Divide $xy = 32$ by $x\sqrt{y} = 16$: $\sqrt{y} = 2 \Rightarrow y = 4,\ x = 8$ **Alternate solution:** Substitute $x = \dfrac{32}{y}$: $\log_2 32 - \log_2 y + \frac{1}{2}\log_2 y = 4$ $5 - \frac{1}{2}\log_2 y = 4$ $\log_2 y = 2 \Rightarrow y = 4,\ x = 8$
23	$x^2 + 2x - 3 > 2x + 3$ $x^2 - 6 > 0$ $(x - \sqrt{6})(x + \sqrt{6}) > 0$ $x < -\sqrt{6}$ or $x > \sqrt{6}$
24	Let $u = x^{\frac{1}{3}}$, so $x^{\frac{2}{3}} = u^2$: $u^2 - 5u + 6 = 0$ $(u - 2)(u - 3) = 0$ $x^{\frac{1}{3}} = 2 \Rightarrow x = 8 \qquad x^{\frac{1}{3}} = 3 \Rightarrow x = 27$
25	$4^{1-x} = \dfrac{4}{4^x}$. Let $u = 4^x$: $u + \dfrac{4}{u} = 5$ $u^2 - 5u + 4 = 0$ $(u - 1)(u - 4) = 0$ $4^x = 1 \Rightarrow x = 0 \qquad 4^x = 4 \Rightarrow x = 1$
26	Let $t = \log_2 x$. Then $\log_x 4 = \dfrac{\log_2 4}{\log_2 x} = \dfrac{2}{t}$: $\dfrac{2}{t} + t = 3$ $t^2 - 3t + 2 = 0$ $(t - 1)(t - 2) = 0$ $\log_2 x = 1 \Rightarrow x = 2 \qquad \log_2 x = 2 \Rightarrow x = 4$ Both are valid bases ($x > 0$, $x \neq 1$).

Q	Solution
27	Pair the outer and inner terms: $n(n+3) = n^2 + 3n$ $(n+1)(n+2) = n^2 + 3n + 2$ Let $m = n^2 + 3n$: $n(n+1)(n+2)(n+3) + 1 = m(m+2) + 1$ $= m^2 + 2m + 1$ $= (m+1)^2 = \left(n^2 + 3n + 1\right)^2$ n is an integer, so $n^2 + 3n + 1$ is an integer, and the expression is a perfect square. **Alternate solution:** Expand both sides: $n(n+1)(n+2)(n+3) + 1 = n^4 + 6n^3 + 11n^2 + 6n + 1$ $\left(n^2 + 3n + 1\right)^2 = n^4 + 6n^3 + 11n^2 + 6n + 1$ These are equal, so the expression is always the square of the integer $n^2 + 3n + 1$.
28 (i)	$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$ $25 = x^2 + \frac{1}{x^2} + 2$ $x^2 + \frac{1}{x^2} = 23$
(ii)	$\left(x + \frac{1}{x}\right)^3 = x^3 + 3x + \frac{3}{x} + \frac{1}{x^3}$ $= x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$ $125 = x^3 + \frac{1}{x^3} + 15$ $x^3 + \frac{1}{x^3} = 110$ **Alternate solution:** Multiply the results: $\left(x + \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2}\right) = x^3 + \frac{1}{x} + x + \frac{1}{x^3}$ $5 \times 23 = x^3 + \frac{1}{x^3} + 5$ $x^3 + \frac{1}{x^3} = 110$

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Q	Solution
29	$y(x-1) = 2x+3$ $xy - y = 2x+3$ $xy - 2x = y+3$ $x(y-2) = y+3$ $x = \dfrac{y+3}{y-2}$ This is undefined when $y = 2$, so no value of x gives $y = 2$. Check: $2 = \dfrac{2x+3}{x-1} \Rightarrow 2x-2 = 2x+3 \Rightarrow -2 = 3$, which is impossible. **Alternate solution:** Rewrite the formula: $y = \dfrac{2(x-1)+5}{x-1} = 2 + \dfrac{5}{x-1}$ $\dfrac{5}{x-1}$ can never be 0, so y can never equal 2.
30	$x^2 - 5x + 6 = (x-2)(x-3)$, so the common root is 2 or 3. Common root 2: $4 + 2k - 6 = 0 \Rightarrow k = 1$. Then $x^2 + x - 6 = (x+3)(x-2)$, roots 2, −3: exactly one in common ✓ Common root 3: $9 + 3k - 6 = 0 \Rightarrow k = -1$. Then $x^2 - x - 6 = (x-3)(x+2)$, roots 3, −2: exactly one in common ✓ $k = 1$ or $k = -1$ **Alternate solution:** At a common root both equations hold. Subtract them: $(x^2 + kx - 6) - (x^2 - 5x + 6) = 0$ $(k+5)x = 12 \;\Rightarrow\; x = \dfrac{12}{k+5}$ This must be 2 or 3: $\dfrac{12}{k+5} = 2 \Rightarrow k = 1$, $\dfrac{12}{k+5} = 3 \Rightarrow k = -1$
31	Let $u = x^2$: $u^2 - (k+1)u + k = 0$ $(u-1)(u-k) = 0 \;\Rightarrow\; x^2 = 1$ or $x^2 = k$ $x^2 = 1$ always gives two solutions, $x = \pm 1$. $x^2 = k$ gives: two new solutions if $k > 0$, $k \neq 1$ (4 in total); no new solutions if $k = 1$ or $k < 0$ (2 in total); one solution $x = 0$ if $k = 0$ (3 in total). So there are exactly three distinct real solutions only when $k = 0$: $x = -1, 0, 1$.
32	$7.2 = \log \dfrac{A_1}{A_0} \qquad 5.8 = \log \dfrac{A_2}{A_0}$ Subtract: $1.4 = \log \dfrac{A_1}{A_0} - \log \dfrac{A_2}{A_0} = \log \dfrac{A_1}{A_2}$ $\dfrac{A_1}{A_2} = 10^{1.4} \approx 25.1$ The main earthquake's amplitude was about 25.1 times larger. **Alternate solution:** $A_1 = A_0 \times 10^{7.2}$ and $A_2 = A_0 \times 10^{5.8}$, so $\dfrac{A_1}{A_2} = 10^{7.2-5.8} = 10^{1.4} \approx 25.1$

Q	Solution
33	$A = A_0\left(\frac{1}{2}\right)^{\frac{t}{5730}}$ $0.18 = 0.5^{\frac{t}{5730}}$ $\frac{t}{5730} = \frac{\log 0.18}{\log 0.5} \approx 2.474$ $t \approx 14\,176$ The animal died about 14 200 years ago. **Alternate solution:** Write $A = A_0 r^t$ with $r^{5730} = 0.5$, so $r = 0.5^{\frac{1}{5730}} \approx 0.999\,879$. $t = \frac{\log 0.18}{\log r} \approx 14\,200$ years
34	Let the other sides be a and b: $a + b = 17$ and $a^2 + b^2 = 169$ $a^2 + (17 - a)^2 = 169$ $2a^2 - 34a + 120 = 0$ $a^2 - 17a + 60 = 0$ $(a - 5)(a - 12) = 0$ The other two sides are 5 cm and 12 cm. **Alternate solution:** $(a + b)^2 = a^2 + b^2 + 2ab \Rightarrow 289 = 169 + 2ab \Rightarrow ab = 60$. a and b add to 17 and multiply to 60, so they are the roots of $t^2 - 17t + 60 = 0$: 5 and 12.
35	The rectangle left over is W by $L - W$. Same shape, so $\frac{L}{W} = \frac{W}{L - W}$ Let $r = \frac{L}{W}$, and divide the right-hand side through by W: $r = \frac{1}{r - 1}$ $r^2 - r - 1 = 0$ $r = \frac{1 \pm \sqrt{5}}{2}$ $r > 0$, so $\frac{L}{W} = \frac{1 + \sqrt{5}}{2}$
36	Income $= p(600 - 20p)$ $p(600 - 20p) \geq 4000$ $20p^2 - 600p + 4000 \leq 0$ $p^2 - 30p + 200 \leq 0$ $(p - 10)(p - 20) \leq 0$ $\$10 \leq p \leq \$20$ (tickets sold are between 400 and 200, so this makes sense).

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Q	Solution
37	If the roots are α and $\frac{1}{\alpha}$, the equation is $(x-\alpha)\left(x-\frac{1}{\alpha}\right) = x^2 - \left(\alpha+\frac{1}{\alpha}\right)x + 1 = 0$ Comparing constant terms with $x^2 - kx + 2k$: $2k = 1 \Rightarrow k = \frac{1}{2}$ With $k = \frac{1}{2}$: $x^2 - \frac{1}{2}x + 1 = 0$, $\Delta = \frac{1}{4} - 4 = -\frac{15}{4} < 0$ So the roots are not real. Ella is not correct: no value of k gives real roots that are reciprocals. **Alternate solution:** α and $\frac{1}{\alpha}$ both satisfy the equation: $\alpha^2 - k\alpha + 2k = 0 \qquad (1)$ $\frac{1}{\alpha^2} - \frac{k}{\alpha} + 2k = 0 \Rightarrow 1 - k\alpha + 2k\alpha^2 = 0 \qquad (2)$ $(2) - (1): \quad (1-\alpha^2) - 2k(1-\alpha^2) = 0 \Rightarrow (1-\alpha^2)(1-2k) = 0$ $k = \frac{1}{2}$ gives no real roots (as above). $\alpha = 1$ (needs a repeated root of 1): $1 - k + 2k = 0 \Rightarrow k = -1$, but then $x^2 + x - 2 = 0$ has roots $1, -2$. × $\alpha = -1$ (needs a repeated root of -1): $1 + 3k = 0 \Rightarrow k = -\frac{1}{3}$, but then the roots are $-1, \frac{2}{3}$. × So Ella is not correct.
38	$8 \times 4^x = 2^3 \times 2^{2x} = 2^{2x+3}$ $x^2 = 2x + 3$ $x^2 - 2x - 3 = 0$ $(x-3)(x+1) = 0$ $x = 3$ or $x = -1$
39	Take $\log_{10}$ of both sides: $\log\left(x^{\log x}\right) = \log(100x)$ $(\log x)(\log x) = \log 100 + \log x$ $(\log x)^2 - \log x - 2 = 0$ $(\log x - 2)(\log x + 1) = 0$ $\log x = 2 \Rightarrow x = 100 \qquad \log x = -1 \Rightarrow x = 0.1$ **Alternate solution:** Let $x = 10^t$, so $\log x = t$: $\left(10^t\right)^t = 100 \times 10^t$ $10^{t^2} = 10^{t+2}$ $t^2 - t - 2 = 0 \Rightarrow t = 2$ or $-1 \Rightarrow x = 100$ or 0.1

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Q	Solution
40	$x^2 - 2kx + 3k = (x-k)^2 - k^2 + 3k$ $(x-k)^2 \geq 0$, so the minimum value is $3k - k^2$ (when $x = k$): $3k - k^2 = 2$ $k^2 - 3k + 2 = 0$ $(k-1)(k-2) = 0$ $k = 1$ or $k = 2$ **Alternate solution:** The minimum is at the axis of symmetry, $x = -\dfrac{-2k}{2} = k$. Substitute: $k^2 - 2k^2 + 3k = 2 \implies k^2 - 3k + 2 = 0 \implies k = 1$ or 2